

Names as Rewards

The Ambiguous Role of Eponyms in the History of Science

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Abstract

Contemporary scientific terminology is populated by thousands of eponyms, technical terms derived from the names of discoverers and inventors. Since the ultimate goals of scientific language are precision and objectivity, the ever increasing use of eponyms is puzzling. Eponyms betray the inherent contradiction between the supposed neutrality and anonymity of scientific discourse and individual vindication. The systematic use of eponyms has resulted in one of the most effective reward systems in Western science. Moreover, it is interesting to note that eponyms do not represent a recent form of recognition; their genealogy can be traced back to classical antiquity, in particular to the Hellenistic era and as a feature of the scientific lexicon they have undergone surprisingly little change over more than two millennia. In my essay I explore the genealogy of scientific eponyms and their cultural background and I shall focus my attention on classical and early modern science. During classical antiquity the use of eponyms served an important dual purpose, on the one hand paying tribute to those individuals who, by dint of their discoveries and written works, gained eminence in a specific field of enquiry, and on the other hand helping to map the historical geography of a discipline and thus establish its boundaries as well as its future prospects. During the Renaissance, with the rediscovery of the classics and the reaffirmation of the cultural value of science and technology, the coining of eponyms gained unprecedented momentum and proliferated in most disciplines.

Keywords

scientific language – scientific eponyms – scientific credit

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Il y a des hommes malheureux. Christophe Colomb ne peut attacher son nom à sa découverte; Guillotin ne peut détacher le sien de son invention.

VICTOR HUGO, *Œuvres complètes* (Bruxelles: Société Typographique Belge, 1837),
p. 603

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An eponym (from the Greek *eponymos*) can refer to many things, from a general methodological prescription (the *Hippocratic oath*) to a geographical region (*America*), a disease (*Boerhaave syndrome*), an anatomical part (*Fallopian tubes*), a mineral (*Berzeliite*), a pharmaceutical remedy (*Mythridatum*), a scientific formula (*Tartaglia's triangle*), a laboratory procedure (the *Pap test*), and experimental practice (*Mesmerism*, *Galvanism*), a scientific law (*Boyle's law*), an instrument or machine (*Archimedes' screw*), a unit of measure (*ampere*, *newton*, *volt*, *watt*, *joule*), or the scientific term for a plant (*Aldrovanda*), and of course a Prize (*Nobel*) which may bear the name of its original discoverer or inventor, or of another eminent scientist deemed to be worthy of commemoration.

Contemporary scientific terminology is populated by thousands of eponyms, technical terms derived from the names of discoverers and inventors. Since the ultimate goals of scientific language are precision and objectivity, the ever increasing use of eponyms is puzzling. How can a scientific term be precise and objective when its root consists of the name of an individual? What exactly does the Pythagorean theorem say about the concept it evokes? Scientific manuals use eponyms without providing their readers with any explanation as to their origins. However, these terms reveal a trait common to all scientific disciplines that deserves to be explored in depth.

The use of eponyms, which has gradually become a characteristic feature of scientific language, seems to be at odds with the objective of the scientific disciplines to express their concepts through a rigorous descriptive terminology. Indeed, one of the major achievements of the natural sciences often seems to be the abstract formalism of their nomenclatures and logic. Even in those disciplines such as botany, which traditionally shunned mathematical rigour, the creation during the 18th century of a highly formalized language was regarded as one of its most important advances. Much earlier the aspiration to rigour lay

at the basis of Aristotle's attempt to impose a formal logic on scientific knowledge. This historical process has led to a degree of specialization that makes it nearly impossible for anyone apart from experts to follow a scientific text in a given field.

However, underlying the structured threads of modern scientific jargon is a linguistic substratum that alludes to an implicit but universal grammar. This grammar, made up in large part of apparently ordinary words, is common to all the sciences and technologies and might even be regarded as a meta-scientific language. Scientists have always been reluctant to talk about eponyms and to provide justifications for their use, but eponymic terms have been accumulating since classical antiquity and now consist of several thousand in every discipline.¹ It is difficult in fact to determine an exact figure for the number of scientific eponyms that have been coined over time because many have for various reasons disappeared, while new ones are constantly, and without fanfare, being devised. The reasons for the survival or disappearance of an eponym is often obscure. Joseph-Ignace Guillotin, the inventor of one of the most famous symbols of the French Revolution, made every possible effort to dissociate his name from this instrument of death but, as we all know, he was irrevocably bound to it in the popular imagination and his heirs were eventually forced to change the family name.

It is probably because of the tacit nature of the knowledge surrounding eponyms that historians of science have not systematically explored what seems to be a fascinating grey zone of scientific practice. However, there has been growing unease regarding the use of eponyms, particularly in the field of medicine, where several articles have appeared expressing doubts about their use in contemporary science. The tide of criticism rose even further after the publication in 2007 of an article entitled "Eponyms in the Nazi Era" in which the authors, both medical doctors, warned that:

Ethical considerations should be introduced into medical nomenclature and nosology just as they exist in patient care and research. This should include the names of individuals who themselves were involved in genocidal atrocities, even if these crimes were not associated with their research of the syndrome after whom they are named. Thus, involvement of such personalities in the deaths of the mentally ill, the medically ill, and racial minorities may be of such enormity as to make it indefen-

¹ It is impossible to estimate with precision the number of eponyms used by scientists. In medicine alone, there are around 20,000 eponyms.

sible to use their names even though their original descriptions may have involved no wrongdoing.²

Although the number of eponyms used by Nazi doctors was relatively small, such repulsive tributes to war criminals demonstrate that this type of name was anything but neutral and could serve purposes totally alien to the principles that one would expect to govern scientific discourse.

In recent times the scientific value of eponyms has been questioned on other grounds as well: (i) they often uncredit other discoverers: the latest case was triggered to Higgs' boson, named after English physicist Peter Higgs. In 2010 an article published in *Nature* by Zeeya Merali pointed out that "the authorship question [was] fraught because the mechanism was developed independently by three groups within a matter of weeks in 1964. First up were Robert Brout and François Englert in Belgium, followed by Peter Higgs in Scotland, and finally Tom Kibble in London, along with his colleagues in the United States, Gerald Guralnik (at the time in London) and Carl R. Hagen";³ (ii) they do not constitute descriptive and meaningful terms unless you are interested in the history of science; and (iii) they emphasize the role of the individual, passing over the collective nature of scientific endeavour. Many contemporary scientists judge eponyms to be an obstacle to the creation of a rigorous and standardized and objective technical language. In commenting on the naming of minerals after distinguished scientists (for example, *biotite*, *gadolinite*, *gaylussite*, *scheelite*), the author of an authoritative textbook on chemical nomenclature⁴ lamented their poor semantics, but chemists and mineralogists have continued to use eponyms. And so do scientists in all other fields. Therefore, despite their myriad shortcomings large and small, eponyms seem to be here to stay.

As suggested by Karl Merton,⁵ the enduring popularity of eponyms might be explained by the central role of originality, a factor that creates a race in which the members of a scientific community are incentivized to compete for recognition as the very first to have made a certain discovery or invention. A

2 Rael D. Strous, Morris C. Edelman, "Eponyms in the Nazi Era: Time to Remember and Time for Change," *Israel Medical Association Journal*, March 2007, 9:207–214, p. 207.

3 Zeeya Merali, "Physicists get political over Higgs," *Nature*, August 4, 2010, <http://doi.org/10.1038/news.2010.390>. More recently it was established that Higgs boson was simultaneously discovered by Peter Higgs and by François Englert and Robert Brout; *Nature*, October 22, 2015, p. 6.

4 Walter E. Flood, *Dictionary of Chemical Names* (Paterford: Littlefield, 1963), p. xxii.

5 Karl Merton, "Priorities in Scientific Discovery: A Chapter in the Sociology of Science," in *The Sociology of Science*, edited by Bernard Barber, Walter Hirsch (New York: The Free Press of Glencoe, 1962), pp. 447–485.

challenge to Merton's hypothesis was raised in 1980 by Stephen M. Stigler in his contribution to Merton's *Festschrift*. The article, which the author with tongue in cheek entitled "Stigler's law of eponymy,"⁶ claimed – on the basis of only a limited number of examples – that "no scientific discovery is named after its original discoverer." Undoubtedly examples can be found of the abuse of eponyms, but Stigler did not quite understand that Merton's interest in the subject extended beyond such misuses and that the ascribing of credit where credit is due does not exhaust the significance of the use of eponyms. Moreover, Stigler did not distinguish between eponyms introduced long after a discovery had been made and those that were coined either when the discoverer was still alive or shortly after his death. This makes a huge difference because, as we all know, scientists are not necessarily accurate historians. Merton therefore concluded that, despite its shortcomings, "Eponymy is the [...] most enduring and perhaps most prestigious kind of recognition institutionalized in science."⁷

In fact, the systematic use of eponyms has resulted in one of the most effective reward systems in Western science. Moreover, it is interesting to note that eponyms do not represent a recent form of recognition; their genealogy can be traced back to classical antiquity, in particular to the Hellenistic era and as a feature of the scientific lexicon have undergone surprisingly little change over more than two millennia. The *longue durée* of this meta-linguistic form of recognising merit probably explains why it has progressively become an automatism transcending the rules of specialized scientific nomenclature. Therefore, it would be enlightening to retrace its history, beginning with the motivations of the ancient and early modern natural philosophers.

1 Scientific Eponyms in Classical Antiquity

Pythagoras inspired several eponyms which, in their earliest phase, associated his name with wisdom and a method of education rather than with a specific invention such as his famous theorem. Plato claimed in the *Republic* that:

Pythagoras was himself especially honored for this (method of education), and his successors, even to this day, *denominating a certain way of life the Pythagorean*, are distinguished among their contemporaries.⁸

6 Stephen M. Stigler, "Stigler's Law of Eponymy," *Transactions of the New York Academy of Sciences*, 1980, 39:147–158.

7 Merton, "Priorities in Scientific Discovery" (cit. note 5), p. 455.

8 Plato, *Republic* X 600b.

About a century later the Greek comic poet Alexis (c. 375–275 BCE) wrote an epigram in which he coined a new term – Pythagorisms – to denote all the “fine-spun discourses, and disputations” that the followers of Pythagoras’ sect indulged in. The near contemporaries Democritus and Hippocrates (both c. 460–370 BCE) were also rewarded with eponyms. A Greco-Egyptian text on magic dating to the 4th century CE and known as the *Leyden Papyrus* contains the description of a “Sphere of Democritus” that could be used calculate whether a patient would recover or die. The text was probably not ascribing the invention of this prognosticating sphere to Democritus himself, but rather using the eponym to refer to the pseudo-epigraphic work of the numerous scientists who in Hellenistic times appropriated his name to augment their own authority.

The towering reputation of Hippocrates in the field of medicine has engendered many eponyms in different periods⁹ but the Hippocratic oath, which remains a central pillar of medical ethics today, is probably the oldest of them all, having been coined in the first century of our era. The authorship of the oath is highly controversial; in 1943 Ludwig Edelstein proposed that it might be traced back to the Pythagorean school.¹⁰ However, Erotian (Herodianus) – a Greek physician working at the court of Nero and the first author to turn his hand to rearranging and ordering the Hippocratic corpus – sought to underscore the achievements of the founding father of medicine by attributing him works of unclear paternity, such as the famous oath. This strategy was quite common at the time and served a specific purpose – to strengthen the authority of medicine as a branch of knowledge. The Hippocratic oath combined the legendary prestige of the Greek physician with what

9 Examples include: “... Hippocratic fingers (in pulmonary osteoarthropathy, the bulbous fingertips that signify chronic lung disease), Hippocratic facies (the face of someone who is in a state of physical shock due to impending death), Hippocratic succussion splashes (shaking a patient in order to hear sloshing, a sign of fluid in the lungs), Hippocratic tumors (malignant melanoma of the skin), and the Hippocrates’ method (redressing an anterior dislocation of the shoulder by applying traction with the hands against the counter-traction created by placing a foot in the patient’s armpit).” Stephen H. Miles, *The Hippocratic Oath and the Ethics of Medicine* (Oxford: Oxford University Press, 2004), p. 31. On medical eponyms see also Eugeniusz Józef Kucharz “Medical Eponyms of Mythological Origin,” *Acta Neophilologica*, 2017, 19/2:29–42, and the master thesis by Carolina Esteban Arrea, *La eponimia en el lenguaje científico. Razones de su existencia y principales problemas que plantea* (Universidad de Salamanca, 2012), https://gredos.usal.es/jspui/bitstream/10366/120785/1/Carolina_Esteban_TFG.pdf (accessed 27 Feb. 2019).

10 Ludwig Edelstein, *The Hippocratic Oath* (Baltimore: The Johns Hopkins University Press, 1943).

were already universally accepted ethical principles of medical practice. The purported authorship underlined the authority of the text, and the authority of the text confirmed the eminence of its author.

The cases of Pythagoras, Democritus and Hippocrates offer interesting examples of eponymic terms introduced in different areas of natural history and emphasize the unequivocal intellectual authority of the individuals in whose honor they were created. And it was precisely during the Hellenistic period, which saw the progressive specialization of the scientific disciplines, that practitioners began to feel the need to anchor their work in a historical context by paying retrospective tribute to the founders and first inventors in their fields.

Regarding botany, for example, Pliny, who discusses eponyms on several occasions, writes:

By Hercules! A single invention before now has elevated men to the rank of gods; and how many an individual has had his name immortalized by being bestowed upon some plant which he was the first to discover, thanks to the gratitude which prompted a succeeding age to make some adequate return.¹¹

But names were not only assigned as a mark of recognition; they could also reflect the personal ambition of the assigner, as Pliny points out:

In former times, there was a sort of ambition, as it were, of adopting plants by bestowing upon them one's name, a thing that has been done before now even by kings, as we shall have occasion to show: so desirable a thing did it appear to have made the discovery of some plant, and thus to have contributed greatly to the benefit of mankind.¹²

One example among the many noted by Pliny is *euphorbia*, a flowering plant widely used in the Roman pharmacopeia that was named by King Juba II after Euphorbos (40–20 BCE), the brother of the emperor Augustus's physician, Antonius Musa. Pliny writes:

Juba II, the father of Ptolemy, who was the first ruler to hold sway over both the Mauretianas, and who is even more distinguished for his renown

¹¹ *Naturalis historia* 25.1.

¹² *Naturalis historia* 25.7.

as a scholar than for his royal sovereignty, has published similar facts about Mount Atlas, and has stated in addition that a plant grows there called the euphorbia, named after his doctor who discovered it; in a volume devoted solely to the subject of this plant he sings the praises of its milky juice in very remarkable terms, stating it to be an aid to clear sight and an antidote against snake-bite and poisons of all kinds.¹³

In addition to plants, eponyms were used to honor the discoverers of pharmaceutical remedies, plasters and pills, especially in ancient Rome. Among them *mithridatium*,¹⁴ the antidote to poisons created by Mithridates VI, the King of Pontus, remained immensely popular in the pharmacopeia until the end of the 18th century.

Pliny's examples show that many of the eponyms introduced in natural history referred to specific discoveries, but the same tendency is clearly detectable in other fields. In a famous epigram by Eratosthenes (c. 276–195/194 BCE), which is cited in Proclus Diadochus' commentary on Euclid's *Elements*, the conic sections (the ellipse, the parabola, and the hyperbola) are defined in tribute to their discoverer as the "triad of Menaechmus."¹⁵ Interestingly, all that is known of Maenechmus's work on conic sections derives from this short epigram. One of ancient Greece's most brilliant astronomers, Meton of Athens (440–410 BCE), is credited with having constructed the mathematical model of an astronomical cycle based on a period of 19 years, leading to the invention of a geared mechanism known as the *antikythera* and to the introduction of a lunisolar calendar that remained the standard for centuries. The cycle soon took on the name of its discoverer. The Greek historian Diodorus Siculus recorded the story as follows:

In Athens Meton, the son of Pausanias, who had won fame for his study of the stars, revealed to the public his nineteen-year cycle, as it is called, the beginning of which he fixed on the thirteenth day of the Athenian month of Scirophorion. In this number of years the stars accomplish their return to the same place in the heavens and conclude, as it were, the circuit of what may be called a Great Year; consequently it is called by some

13 *Naturalis historia* 5.16. See also 25.77–79.

14 See *Naturalis historia* 23.149, 25.6, and 29.24.

15 Eratosthenes wrote "Don't produce the conic section with the triads of Menaechmus" (the triad being made up of the parabola, hyperbola and ellipse). Menaechmus (365–325 BCE) was a pupil of Eudoxos and an associate of Plato. The citation is from Proclus' commentary on Euclid (111, 23).

the Year of Meton. And we find that this man was astonishingly fortunate in this prediction which he published; for the stars complete both their movement and the effects they produce in accordance with his reckoning. Consequently, even down to our own day, most Greeks use the nineteen-year cycle and are not cheated of the truth.¹⁶

Engineers and the inventors of other ingenious mechanical devices also had the honor of an eponym bestowed upon them. Again, Diodorus Siculus praised one of them by recounting an interesting episode from his career:

... and it was these men who quarried the stones of which not only the largest temples of the gods were constructed but also the underground conduits that were built to lead off the waters from the city [Agrigento]; these are so large that their construction is well worth seeing, although this is little thought of since they were built at slight expense. The builder in charge of these works, who bore the name of Phaeax, brought it about that, because of the fame of the construction, the underground conduits received the name "Phaeaces" from him.¹⁷

The Romans honoured also discoverers. Pliny thus reported the discovery of obsidian, a stone which immediately became immensely popular throughout the Empire:

among the various forms of glass we may reckon Obsian glass, a substance very similar to the stone found by Obsius in Ethiopia.¹⁸

Although we do know nothing about Obsius, his discovery has made him famous throughout ages.

The Romans were even disposed to honor inventions made in the minor crafts, as in the case of the introduction of a new type of garment:

A Roman garment [was named] after Tebennos, the Arcadian who first donned this outer-garment, for he sailed down the Ionian Gulf and was received by the settlers living there. [And] learning from him, the natives clothed themselves in the same way and called the garment a *teben-*

¹⁶ Diodorus Siculus, 12.36.2–3.

¹⁷ Diodorus Siculus, 11.25.3–4.

¹⁸ *Naturalis historia* 35.196.

neion, named after the inventor. But subsequently the name changed and became *tebennos*.¹⁹

During classical antiquity the use of eponyms served an important dual purpose, on the one hand paying tribute to those individuals who, by dint of their discoveries and their written works, gained eminence in a specific field of enquiry, and on the other hand helping to map the historical geography of a discipline and thus establish its boundaries as well as its future prospects. It is not surprising to find that the introduction of eponyms in the scientific disciplines coincided with the publication of collections of biographies of scientists aimed at honouring their achievements. In this phase, during the Hellenistic era, eponyms marked critical achievements in the history of science and technology and probably also served to encourage emulation by up and coming scientists.

2 Eponyms Conquer the Universe

During the Renaissance, with the rediscovery of the classics and the reaffirmation of the cultural value of science and technology, the coining of eponyms gained unprecedented momentum and proliferated in most disciplines. It is well known that the naming of America (Fig. 1) can be retraced to a specific year, 1507, and a specific scientific work – Martin Waldseemüller's cosmography²⁰ – in which the author justified his choice as follows:

But now these parts have been more widely explored, and also another fourth part has been discovered by Americus Vesputius (as will be heard in the following), and I do not see why anyone should justifiably forbid it to be called Amerige, as in "Americus' Land," or America, from its discoverer Americus, a man of perceptive character; since both Europe and Asia have received their names from women.²¹

The 'wrong' attribution of the geographical 'discovery' of America is interesting for many reasons, but here it suffices to mention that the scientific background of Vespucci and his active interest in perfecting instruments of navigation must

19 *Suda*, 465.

20 Martin Waldseemüller, *Universalis cosmographia secundum Ptholomaei traditionem et Americi Vespucii alioru[m]que lustrationes* (St. Dié, 1507).

21 Martin Waldseemüller, *Cosmographiae introductio* (St. Dié, 1507), n.p.



FIGURE 1
Detail of Martin Waldseemüller,
Universalis Cosmographia
ST. DIE, 1507



FIGURE 2
The portrait of Amerigo Vespucci in
Martin Waldseemüller, *Universalis Cos-*
mographia
ST. DIE, 1507

have played a significant role in Waldseemüller's decision to give his name precedence over that of Christopher Columbus. It is certainly not by chance that in the map accompanying the book, Vespucci is portrayed facing Ptolemy and holding a compass, thus emphasizing the crucial role played by science in geographical discoveries. (Fig. 2) Waldseemüller's eponym is interesting in another respect; by honouring an eminent Florentine, the German cosmographer underlined the innate cosmopolitism of scientific endeavour, thus ranking science on the same level as other universal forms of knowledge such as philosophy and humanism. However, in this case eponyms added to universalism the value of progress, which manifested itself by the exploration and discovery of new worlds.

During the early modern epoch, eponyms continued to proliferate in medicine, mathematics, alchemy and pharmacy, and paid tributes to a new generation of natural philosophers. In astronomy Galileo commemorated Jupiter's newfound moons by calling his discovery *Medicea Sidera* in honor of his future patrons, the Medici, (Fig. 3) although he refrained from bestowing similar accolades on his peers. With the rapid progress made in telescopic observations, it was not long before the craters, mountains and regions of the moon were discovered and baptized with the names of ancient and modern astronomers.²² Interestingly, the most popular and enduring work in this regard was *Almage-*

22 Ewen A. Whitaker, *Mapping and Naming the Moon: A History of Lunar Cartography and Nomenclature* (Cambridge: Cambridge University Press, 1999).



FIGURE 3

Medal by Gasparo Mola (1615) portraying Jupiter holding thunderbolts in his right hand and a scepter. On his right an eagle and around him the Medicean stars discovered by Galileo. The medal was dedicated to Cosimo II de Medici

tum novum, astronomiam veterem novamque complectens (Bologna, 1651) by the Jesuit priest and astronomer Gian Battista Riccioli, which contained a detailed map of the moon whose features, in keeping with the new eponymic nomenclature, were labelled with the names of the following ancient, medieval and modern astronomers (Figg 4–5):

Anaximenes, Anaxagoras, Archimedes, Architas, Aristarchus, Aristotle, Berosius, Cleomedes, Democritus, Ecfantus, Erathostenes, Eudoxos, Geminus, Heraclides, Hipparchus, Macrobius, Manilius, Meton, Philolaus, Pliny the Elder, Plutarch, Posidonius, Proclus, Ptolemy, Pythagoras, Seneca, Sosigenes, Sulpicius Gallus, Thales, Timaeus, Vitruvius, Xenophanes, Zoroaster,

Apianus, Bessarion, Boulliau, Cardano, Campano da Novara, Cavalieri, Clavius, Copernicus, Cusanus, Fernel, Finé, Fracastoro, Galileo, Gassendi, Gemma Frisius, Grimaldi, Hevelius, Kepler, Kircher, Lansbergius, Magini, Manzini, Maurolico, Regiomontanus, Rheticus, Riccioli, Tycho, Scheiner, Snell, Stevin.

Riccioli's map also coined eponyms that commemorated the inventions of various instrument makers, such as Francesco Fontana and Carlo Maria Manzini, thus providing a balanced overview of a science that, with the introduction of the telescope, began to rely more and more on the skills of craftsmen.

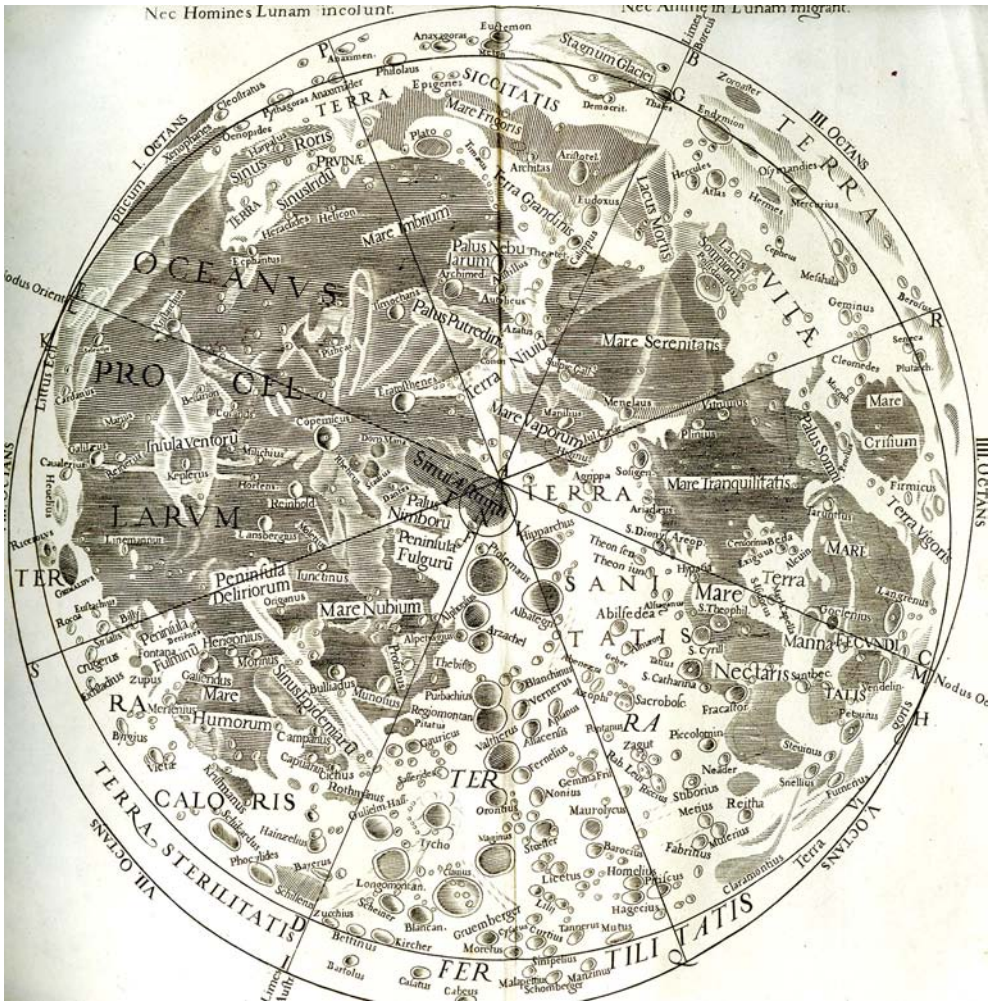


FIGURE 4 The image of the moon in Riccioli's atlas

The success of *Almagestum novum* inspired subsequent astronomers, who continued the practice of using eponyms to honor distinguished predecessors.

The success and visual impact of the eponyms introduced in cosmography and astronomy generated emulation in all the other sciences. It was in fact during the sixteenth and seventeenth centuries that anatomical eponyms were introduced²³ and in other sciences, such as botany,²⁴ chemistry and pharma-

23 Kurt O. Gilliland, Royce L. Montgomery, *Anatomists and Eponyms* (Nottingham: Nottingham University Press, 2011).

24 Philippe Selosse, "La théorie de l'éponymie en botanique à la Renaissance," in *Ecriture du*



FIGURE 5 Detail of Riccioli's moon atlas with eponyms

cology rediscovered. While the use of scientific eponyms rapidly grew among early modern authors, it is extremely rare to find a text containing a theoretical justification of their presence.

The Swedish naturalist Carl von Linné introduced the systematic use of eponyms when he formalized the modern, binomial system of botanical nomenclature and he was among the first to explain their use.²⁵ On this, in 1737 he declared:

When men have devoted a happy life to study in other branches of Science, and have accomplished something noteworthy, their names are customarily attached to their accomplishment: so is it with Physicians, Anatomists, Pharmacists, Chemists, Surgeons, etc. Hence we hear of *Harveian circulation*, the *Glands of Nuck*, ... the *Torricellian Tube*, the *Pills of Fracastoro*, ... etc. Why then should we deny to Botanists all reward, all honour, all commemoration?²⁶

texte scientifique au Moyen Âge, edited by Claude Thomasset (Paris: Presses de l'Université de Paris-Sorbonne, 2006), pp. 265–278.

25 Ilva Beretta, "Italian Naturalists in Linnaeus' Herbarium," in *Linnaeus in Italy. The Spread of a Revolution in Science*, edited by Marco Beretta, Alessandro Tosi (Sagamore Beach: Science History Publications/USA, 2007), pp. 1–14.

26 Carl Linnaeus, *The "Critica Botanica" of Linnaeus. Translated by Sir Arthur Hort* (London: John Ray Society, 1938), p. 64.

In the eyes of Linné, it was unjust to dismiss botany as a minor science, because the efforts displayed by botanists in the exploration of nature were second to none and deserved renewed consideration:

Alas that such labour should go unrewarded! Would you not like, dear Botanists, to have some reward within your reach which you can obtain by merit, and with which you might rouse the ambition of distinguished men?²⁷

In coining hundreds of botanical eponyms, Linné was careful to distinguish those that paid tribute to an eminent classical or Renaissance author from those rewarding contemporary botanists who had earned distinction by discovering a new specimen. Ever since Linné's reform, the nomenclature of botany has been largely dominated by his and his successors' systematic use of eponyms. While Linné's binomial nomenclature was regarded during the 18th century as a major achievement of modern science that could be applied to other disciplines, his use of eponyms instead evokes the image of science that was centered around authority, a notion that many authors of the Enlightenment were beginning to oppose.

The case of Linné in fact demonstrates that through the sheer force of one's personal authority, a prominent scientist might use eponyms as an instrument for arbitrary discrimination against his opponents. And indeed we find that the eminent Swede did not always employ eponyms in an objective and constructive manner, one case in point being *Bufonia*, an unimposing plant with a disagreeable smell that he named after George Leclerc de Buffon, the fiercest critic of his reform of the botanical nomenclature and, indirectly, of the use of eponyms in science. While they could be used to reward originality and distinction in scientific work, eponyms could also be exasperatingly ambiguous and many thought they should be entirely eliminated from the lexicon of science. However, the removal of such entrenched terminology could at no point have been an easy task.

The subterranean power exerted by eponyms in science can be illustrated with a famous example of the reform of the technical language drawn from the history of chemistry. In 1787 the French chemist Antoine Laurent Lavoisier together with a small number of associates published a succinct volume entitled *Méthode de nomenclature chimique*. In it the authors proposed a comprehensive reform of the chemical nomenclature and the introduction of a

²⁷ Ibid., p. 69.

method for naming chemical substances, the rigor of which they hoped would be as accurate as that of algebra. Their task was all the more ambitious because the traditional chemical nomenclature coincided in large measure with that of pharmacy, which was regulated throughout Europe by the frequent publication of official municipal pharmacopoeias. In order to justify their linguistic revolution Lavoisier and his team pointed out the ambiguity and obscurity of the old chemical names, many of which were eponyms. To Lavoisier the names of substances that were derived from the names of their first discoverers – Glauber's salt, Delhuyar's acid, Scheele's fire air, Lilium of Paracelsus, Boyle's fuming liquor and many more – made little sense for two reasons: (i) they did not convey the chemical principles underlying the composition of these substances, and (ii) they recounted a history that embraced alchemy, which Lavoisier wished to cancel from the horizons of the new science of chemistry.

In less than twenty years Lavoisier's nomenclature was adopted throughout Europe and the old names disappeared from the language of chemistry, although a handful of historic eponyms managed to survive. One traditional name – Glauber's salt – celebrated the discovery of sodium sulphate by the able German alchemist, Johann Rudolf Glauber, and was still in use at the beginning of the 20th century. What is more, it did not take long before new eponyms were being introduced to denote new substances, instruments, and reactions. To cite just two examples: in 1802 the French chemist François Descroizilles published a booklet entitled *Description et usage du Bertholimètre*, in which he described an instrument to analyse alkali compounds that was named after Claude Louis Berthollet, one of the associate authors of Lavoisier's *Méthode de nomenclature chimique*. The periodic table itself includes 14 elements that were named after distinguished scientists who were not necessarily their discoverers:

curium (Cm, 96) – Pierre and Marie Curie
einsteinium (Es, 99) – Albert Einstein
fermium (Fm, 100) – Enrico Fermi
gallium (Ga, 31) – Lecoq de Boisbaudran
hahnium (105) – Otto Hahn
lawrencium (Lr, 103) – Ernest Lawrence
meitnerium (Mt, 109) – Lise Meitner
mendelevium (Md, 101) – Dmitri Mendeleev
nobelium (No, 102) – Alfred Nobel
roentgenium (Rg, 111) – Wilhelm Roentgen (formerly Ununumium)
rutherfordium (Rf, 104) – Ernest Rutherford
seaborgium (Sg, 106) – Glenn T. Seaborg

The popularity of chemical eponyms not only survived the linguistic revolution wrought by Lavoisier, but have flourished with even stronger vigour.²⁸

Eponyms reflect a paradoxical feature of science; on the one hand they were intended to celebrate the historic achievements of eminent scientists, but on the other hand they undermine the objectivity that is supposed to be the main characteristic of scientific nomenclature by emphasizing the role of individuals. Eponyms in fact betray the inherent contradiction between the supposed neutrality of scientific discourse and individual vindication, and illustrate both the importance and the limits of the reward system in science.

28 Kurt G. Wagner, *Autoren-Namen als chemische Begriffe* (Weinheim: Verlag Chemie, 1951).